

Emerging Trends in Computer Science (CSC211S2)

Assignment on IoT

Overview:

This is a group assignment with individual marking components. Each Group can have a maximum of 5 members. The task is to conceptualize and develop an IoT project idea with sufficient sophistication.

IoT Project Requirements:

- The project must incorporate the following capabilities:
 - At least one sensing component.
 - At least one controlling/actuator component.
 - Communication capabilities.
 - Consider security aspects.
- Simulate the project components using an online CAD tool (e.g., Tinkercad). Each member must complete their simulation individually, and this will be marked separately.
- Build a working prototype of the IoT solution using actual Arduino, sensors, and actuators (group activity). You can select the best simulation from the members and build a prototype.

Expected Outputs: (Deadline: 5th Septemebr 2025 at 4 pm)

- Digital Poster (Group Output):
 - A single-page digital poster highlighting your project idea, implementation, and significance.
- Simulation or Sketch (Individual Output):
 - Each member must create and present their simulation using an online CAD tool (e.g., Tinkercad).
- Group Demo:
 - Showcase the working prototype with Arduino, sensors, and actuators.
- VIVA:
 - Individual: Focused on each member's contribution.
- 15 minutes for Group allocated for Demo and VIVA

The project must reflect creativity, technical feasibility, and an understanding of IoT principles.

IoT Project Evaluation Rubric

Criteria	Excellent (10-9)	Good (8-7)	Satisfactory (6-5)	Needs Improvement (4 or below)	Weight
Creativity and Innovation	Project demonstrates a highly original and innovative idea with clear problem-solving intent.	Idea is original and somewhat innovative, with good problem-solving aspects.	Idea is moderately creative but lacks significant innovation.	Idea is generic and lacks creativity or innovation.	10%
Project integration and features	All components (sensing, controlling, communication) are used and integrated seamlessly.	Most components are functional, with minor integration issues.	Some components are functional, but integration is incomplete.	Components are mostly non-functional or poorly integrated.	10%
Technical Complexity	Project demonstrates advanced technical complexity, exceeding expectations for the course level.	Project shows good technical complexity, meeting the course expectations.	Project demonstrates basic technical complexity, with room for improvement.	Project lacks technical depth and sophistication.	5%
Other Considerations	Comprehensive security measures are implemented effectively in the design and prototype.	Adequate other measures are implemented with minor gaps.	Minimal security measures are considered, with significant room for improvement.	Security aspects are poorly addressed or missing.	5%
Simulation (Individual)	Simulation is complete, accurate, and well-presented with proper use of the CAD tool.	Simulation is mostly accurate with minor issues in presentation or completeness.	Simulation is functional but lacks accuracy or clarity in presentation.	Simulation is incomplete or poorly executed.	20%
Prototype Development (Group)	Prototype is fully functional and demonstrates a clear understanding of IoT principles.	Prototype is mostly functional, with minor flaws in implementation.	Prototype is partially functional, with noticeable issues in implementation.	Prototype is non-functional or poorly implemented.	20%
VIVA (Individual)	Digital poster is good, very good understanding, and engaging.	Good understanding, with minor lapses in clarity.	Understanding is moderately fine but lacking in some areas.	VIVA is poor and unclear or lacks engagement in the project.	30%